Tadikonda Sai Chakrri

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EDUCATION

Indiana University Bloomington, Luddy School of Informatics, Computing and Engineering — ***August 2025 - May 2027***
Master of Science in Computer Science — GPA: 3.83

Vellore Institute of Technology — ***September 2020 - September 2024***
Bachelor of Technology in Computer Science and Engineering — GPA: 8.70

TECHNICAL SKILLS

Programming	Java, Python, R Language, React (NextJs), C Programming
Frameworks	Flask, TensorFlow, Scikit-Learn, NumPy, Pandas
Tools & Platforms	AWS, Snowflake, Jupyter, Figma, Selenium, PySpark
Databases	Neo4j, MySQL, Firebase, Pinecone, Mongo DB, Chroma DB
AI/ML	RAG, Prompt Engineering, LLM & Agent Evaluation, MLOps, OpenCV

EXPERIENCE

AI Learning & Instructional Specialist — ***July 2026 - Present***
O'Neill School of Public and Environmental Affairs, IU — *Bloomington, IN*

- Built **AI agents** and **agentic workflows** for graduate-level project management class, enabling students to explore project scenarios, and develop practical recommendations through iterative work with LLMs.
- Designed hands-on GenAI learning activities that guided students from writing effective prompts to using multi-step AI workflows for project planning, analysis, and decision-making.

Research Assistant — ***April 2026 - Present***
School of Public Health, IU — *Bloomington, IN*

- Conduct large-scale **epidemiological** and **statistical** analyses across SWAN, DHS, and NHANES datasets to investigate diabetes, nutrition, anemia, weight, and cardiometabolic biomarkers, transforming complex health data into reproducible research insights.
- Build reproducible **data pipelines** for data cleaning, harmonization, **feature engineering**, survey-weighted and longitudinal modeling, sensitivity analysis, visualization, and manuscript validation using statistical methods.

AI Systems Engineer (GenAI) — ***January 2025 - July 2025***
Tata Consultancy Services — *Bangalore, India*

- Developed a Retrieval-Augmented Generation based **AI** solution using **ChromaDB**, **Streamlit** and an internal LLM on the Lumin platform.
- Created an end-to-end data pipeline achieving about **86%** retrieval precision initially and cutting down response latency by at least **15%** through prompt optimizations and secure vector search.
- Collaborated cross-functionally with **Data scientists**, **Product owners**, and **Domain experts** to gather requirements, align deliverables, and communicate complex technical concepts clearly enhancing stakeholder satisfaction and accelerating adoption of the AI solution.

Full Stack Developer Intern (AI) — ***August 2024 - December 2024***
Smallest.ai — *San Fransico, CA (Remote)*

- Engineered a scalable **Fast-API**-based system that integrated the company's proprietary speech synthesis API. This system enabled the seamless conversion of text files to podcast-style audio, achieving high accuracy of **88%** audio text alignment.
- Initiated an **AI enhanced** document processing pipeline that leveraged natural language processing techniques to automatically clean, summarize and utilized smallest's inhouse models to vocalize long form text.
- This innovative solution resulted in a **7%** increase in user retention during beta testing by providing an engaging, human-like audio podcast experience.

Data Analyst Intern — ***August 2022 - November 2022***
Livspace — *Bangalore, India*

- Built end-to-end analytics workflows to clean, normalize, transform, and analyze large operational datasets, translating business and marketing data into interactive dashboards, KPI visualizations, and actionable insights used by the analytics Operations team to optimize internal processes and support targeted marketing and strategic decision-making.

PROJECTS & RESEARCH WORK

DocFactory: Document Intelligence & Extraction Platform — **GitHub/DocFactory**

- Built a **multi-tenant** AI document intelligence platform that ingests invoices and automatically parses, extracts, and validates structured data through an asynchronous validation pipeline. Integrated MinIO, ElasticMQ, automated testing and evaluation pipelines for scalable, production-ready document processing.

Visual-AI-ze: A Mathematical Visualization System — **GitHub/VisualAIze**

- Engineered a **multi-agent LLM visualization** system using RAG and **agent orchestration** to transform mathematical topics and research documents into 2-to-4-minute animated explanations, automating document parsing, concept mapping, pedagogical planning, scene design, and code generation with Manim and reducing manual visualization effort.

MemOS: Memory Operating System with Knowledge Graph — **GitHub/MemOS**

- A full-stack AI memory system combining BM25 retrieval, LLM-powered cited Q&A, knowledge graphs, and FSRS 4.5 to transform research into provenance-aware atomic memories with cross-source linking, scalable ingestion, retrieval, visualization, and persistent storage.

ROLES & RESPONSIBILITIES

GHI Project	**Data Analyst** at Global Health Impact, analyzing global disease trends & treatment accessibility across decades using large scale health datasets to evaluate the burden of infectious diseases, improve data driven public health insights.
Project 990	**Computer Scientist** at Project 990, developing data driven tools to analyze millions of nonprofit records (Form 990), enabling insights into philanthropic trends, funding patterns, and organizational performance.
GPSG	**Student Representative** at the Graduate Professional Student Government IU, advocating for peers, facilitating access to research funding opportunities, and representing student concerns at departmental and university levels.